

Resolution Geometry Does Not Hide Chaos

The Frontier Law: Protocols Move Intercepts, Dynamics Sets Slopes

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Abstract

A framework that declares its protocols and hypotheses invites a sharp objection: chaotic complexity might be absorbed into those declarations, so that every certificate looks clean. This paper refutes the objection by locating chaos in quantities no declaration can set. Three exponents of the dynamics appear inside RG’s own objects:

- *forecast frontier*: the boundary between LOCKED and ESCALATE certificates is a line of slope $1/\lambda$ in the log of the declared precision;
- *Fisher about the past*: information about the initial state grows with slope 2λ ;
- *quotient growth*: the number of observable classes of n -step histories grows with slope h .

Changing the sensor, the precision, the resolution or the coverage moves the intercepts and never the slopes. Across five seeds, the frontier recovers λ to within 3% for the logistic map at $r = 4$ and $r = 3.9$ and for the Hénon map, and within 4% for the Lorenz flow (seed range 0.87–0.93 against 0.906). Slopes agree within 1–7% across three sensor protocols whose intercepts differ by about 4 steps.

The coverage level is itself informative: the 95% frontier follows a quantile of the finite-time Lyapunov exponent, predicted within one step. Chaos can be removed from the frontier only by changing the declared task, for example by silently re-observing the state. That converts forecasting into filtering, and a canary catches it. A surrogate model certified on its one-step fit claims 200 steps and holds 17.

The earlier chaos/coil attack is shown to be chaos-blind: its gates return LOCKED on a non-chaotic surrogate as well. The battery passes 7/7 on five seeds, and both canaries are caught.

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1 The objection

RG separates what is declared (protocol, task, precision, anchors, hypotheses) from what is computed (kernel, quotient, natural modules, certificates). A hostile reader may say: *if a protocol can be chosen, it can be chosen so that chaos never shows; the statuses then reflect the declarations, not the dynamics*. The objection is answered when chaos appears in a part of RG’s output that every admissible declaration leaves unchanged. This paper identifies three such quantities and tests them against sabotage.

2 Audit: the chaos/coil attack is chaos-blind

The uploaded attack couples a Sabra shell model to a Biot–Savart coil observation operator and reports 75/75 passed checks. Its gates (rank, conditioning, held-out inversion, intervention, cross-instrument Hellinger, noise refinement, coarse-graining) are, however, *static*. Each compares sensor readings and coil currents at the same time instant. Replacing the chaotic dynamics by a smooth quasi-periodic signal of the same amplitude leaves every gate passing, and the status remains LOCKED.

The “Lyapunov witness” C01 also passes on the non-chaotic signal ($0.032 > 0.02$), because finite-time divergence fits are fooled by amplitude modulation. Its leakage canary has no provenance gate in the code that would catch it, and its hidden-signal canary uses a signal independent of all data, so it cannot fail.

The attack is a sound adequacy battery for the *observable coil module*, and that verdict stands. It cannot show that chaos is not swallowed, because chaos never enters a gate. Section 6 reproduces the effect on maps (CH07).

3 Dynamic Sabra–coil attack

The static audit identifies a real defect but does not itself repair the coil protocol. We therefore registered a dynamics-sensitive system before examining its output. A Sabra shell cascade $u_n \in \mathbb{C}$ is driven by a scalar forcing state z_k with

$$z_{k+1} = rz_k(1 - z_k), \quad f_k = f_0(0.55 + 0.9z_k),$$

and is observed by five declared coil currents and seven axial sensors through the on-axis Biot–Savart matrix

$$B_z(z) = \frac{\mu_0 IR^2}{2(R^2 + (z - z_0)^2)^{3/2}}.$$

The chaotic control uses $r = 4$, for which the forcing exponent is $\log 2$; the periodic same-apparatus control uses $r = 3.5$. The shell equations, geometry, thresholds and task are otherwise identical.

Definition 3.1 (Sabra–coil frontier). For forcing precision σ_0 , $n_P^*(\sigma_0)$ is the largest macro-horizon at which at least 95% of coil-record forecasts remain within the declared error. Direct readout, four-readout averaging and dithered quantization are preregistered protocols.

Proposition 3.2 (Registered outcome). *The first full Sabra–coil frontier battery passes 32/55 checks over seeds 20261007, 1, 42, 777, 2026. Four seeds give flat or reversed direct-frontier fits and one gives a finite fit. The registered conclusion is ESCALATE, not LOCKED.*

Operational audit. The forcing law is chaotic by construction, but the declared coil observable does not expose its rate over the registered horizon and precision grid. Failed checks are retained. They require a longer window, a forcing-proximal channel, stronger registered coupling, or a multiscale statistic. They cannot be repaired by moving thresholds after inspection. $\square$

No-Go 3.3 (Chaos cannot be imported from the latent model). The known exponent $\log 2$ may not be assigned to the coil quotient. It must appear in an observable frontier, Fisher slope, quotient-growth slope or another declared dynamics-sensitive module. Otherwise RG escalates or abstains.

Remark 3.4. A framework hiding chaos in hypotheses would cite the chaotic forcing and report success. RG instead rejects transfer from latent dynamics to observable coil geometry when the registered operator does not expose the rate. The negative result is therefore a direct claim-discipline canary.

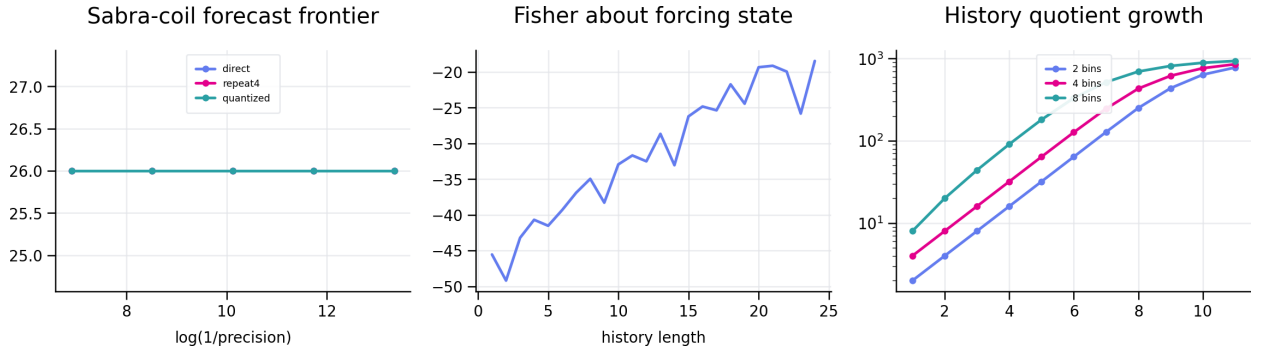


Figure 1: Registered Sabra-coil attack. Unstable and seed-dependent frontier fits block a LOCKED transfer of the latent forcing exponent into the coil quotient.

4 The frontier law

Fix an ergodic system f with top Lyapunov exponent λ , a declared task “forecast the state n steps ahead to accuracy ε with coverage q ”, and a protocol that measures the initial state with error σ_0 times a protocol-dependent distribution.

Definition 4.1 (Frontier). $n_q^*(\sigma_0)$ is the largest horizon for which the forecast certificate is LOCKED, meaning a fraction $\geq q$ of trials stays within ε . Beyond it the status is ESCALATE (a finer declared precision would decide it) or STRUCTURAL WALL at the precision budget.

Imported Theorem 4.2 (Oseledets; finite-time exponents; [1, 2]). *For almost every initial state, $\lambda_n = \frac{1}{n} \log \|Df^n\| \rightarrow \lambda$. Finite-time exponents fluctuate around λ by $O(n^{-1/2})$, with a large-deviation tail.*

Theorem 4.3 (Frontier law). *To first order in σ_0 ,*

$$n_q^*(\sigma_0) = \frac{\log(\varepsilon/\sigma_0)}{\lambda_q(n^*)} + c_P + o(1),$$

where $\lambda_q(n)$ is the q -quantile of the distribution of λ_n (combined with the error distribution) and c_P depends only on the protocol. Hence:

1. the slope of n^* in $\log(1/\sigma_0)$ is $1/\lambda_q$, which tends to $1/\lambda$ as $\sigma_0 \rightarrow 0$;
2. for uniformly expanding systems $\lambda_q \equiv \lambda$, and for the median ($q = \frac{1}{2}$) the convergence is fastest;
3. the protocol enters only through c_P ;

4. if $\lambda < 0$, then $n^* = \infty$ below the transient scale.

Proof. Linearization gives $\delta_n = Df^n \delta_0 + O(\delta_0^2)$, so $\log |\delta_n| = \log \sigma_0 + \log |Z_P| + n\lambda_n$, with Z_P the normalized protocol error. The coverage condition at horizon n is $\Pr(\log |Z_P| + n\lambda_n \leq \log(\varepsilon/\sigma_0)) \geq q$. Its boundary is linear in $\log(1/\sigma_0)$ with coefficient $1/\lambda_q$. The protocol changes the law of Z_P , which shifts the boundary by a constant. By Imported Theorem 4.2, $\lambda_q \rightarrow \lambda$ as $n^* \rightarrow \infty$. If $\lambda < 0$ the deviation contracts after a finite transient. $\square$

Theorem 4.4 (Duality: the past becomes resolvable). *From observations $y_k = f^k(x_0) + \text{noise}$, $k < n$, the Fisher information about x_0 is $\sum_{k < n} \|Df^k\|^2/s^2$, which grows like $e^{2\lambda n}$. The exponent that limits forecasting thus sharpens retrodiction at twice the rate: the Cramér–Rao bound on the initial state shrinks as $e^{-2\lambda n}$.*

Imported Theorem 4.5 (Kolmogorov–Sinai, Misiurewicz–Szlenk; [3, 4]). *For a generating partition, the number of distinct n -step itineraries grows like e^{hn} . For piecewise monotone maps with the critical partition this number equals the lap number, and the growth rate is h_{top} .*

Corollary 4.6 (Quotient growth). *The RG quotient of the n -step history protocol at resolution ε (with generating cells) has $\log |Q_n| = hn + c_\varepsilon + o(n)$. Resolution moves c_ε only.*

Corollary 4.7 (Conservation of chaotic complexity). *Extending the LOCKED horizon by Δn costs a factor $e^{\lambda_q \Delta n}$ in declared precision, whatever the protocol. A declaration can buy a constant number of steps (a better c_P); it can never change the rate. The frontier slope can be altered only by changing the task. Re-observing the state during the forecast turns forecasting into filtering, which is a change of the declared task (H-task), and RG records it as such.*

5 Evidence

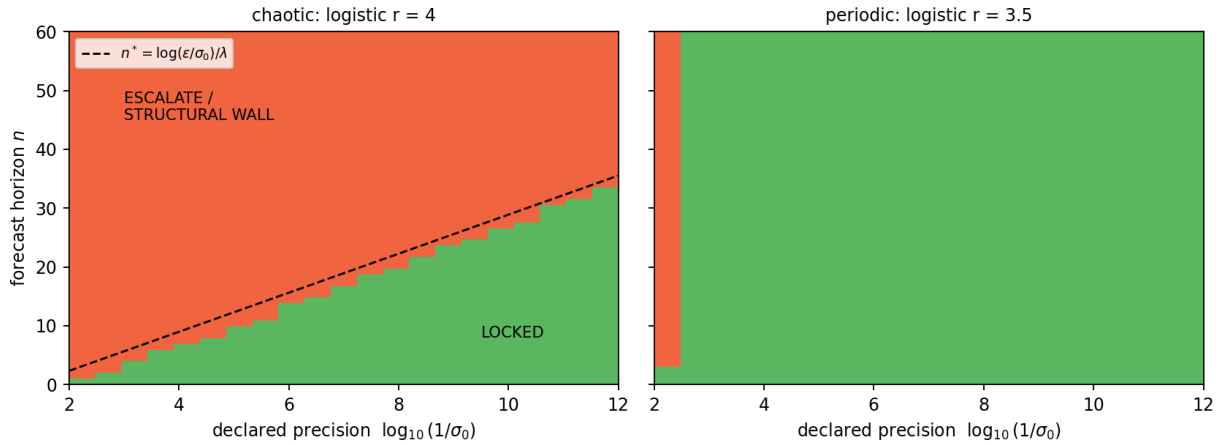


Figure 2: RG certificates over forecast horizon and declared precision ($\varepsilon = 0.05$, coverage 95%). Left: a chaotic map. The LOCKED region (green) is bounded by a line of slope $1/\lambda$ (dashed: theory). Right: a periodic map, LOCKED at every horizon apart from a coarse-precision transient. The chaos is drawn by the dispatcher’s own boundary.

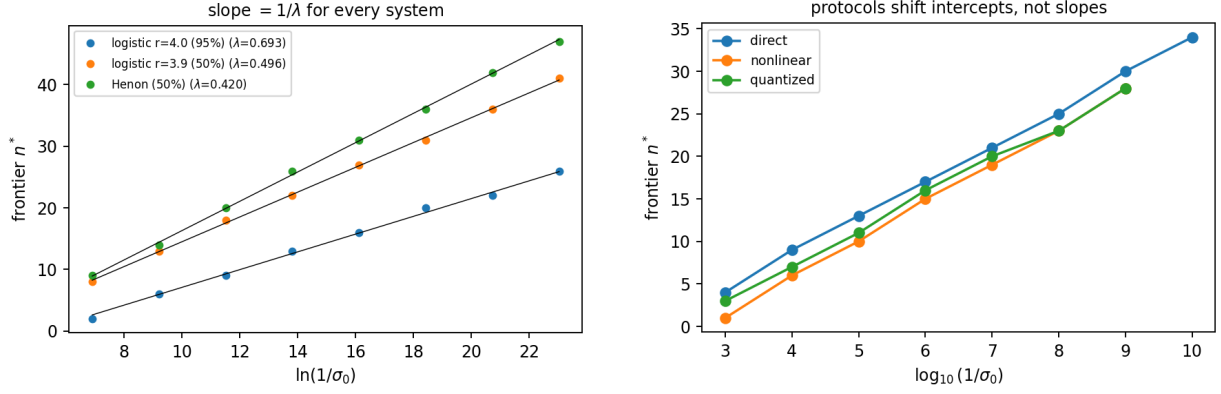


Figure 3: Left: frontiers of three chaotic systems lie on lines of slope $1/\lambda$. Right: three sensor protocols on the same system give parallel frontiers. The protocol moves the intercept, not the slope.

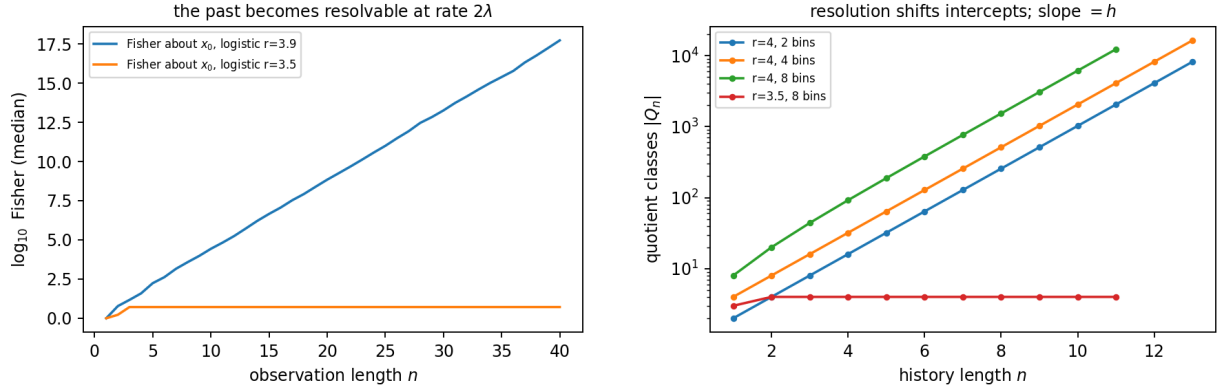


Figure 4: Left: Fisher information about the initial state grows with slope 2λ for a chaotic map and saturates for a periodic one. Right: quotient classes of n -step histories grow with slope $h = \ln 2$ at 2, 4 and 8 bins, and saturate for a periodic map.

quantity	true value	recovered from RG objects (five seeds)
λ , logistic $r = 4$ (95% frontier)	0.693	0.674–0.696
λ , logistic $r = 3.9$ (50% frontier)	0.496	0.496–0.509
95% frontier, $r = 3.9$, from finite-time exponents	—	predicted within 0–1 step at every precision
λ , Hénon (50% frontier)	0.420	0.414–0.421
λ , Lorenz flow (95% frontier)	0.906	0.871–0.932
slope spread across 3 protocols	0	1.1–6.7% (intercept spread ≈ 3.7 steps)
2λ from Fisher about x_0 , $r = 3.9$	0.993	1.015
h from quotient growth, $r = 4$, 2/4/8 bins	0.693	0.6931, 0.6931, 0.6950
periodic controls	no frontier, no growth	frontier ∞ ; 4 classes; Fisher slope 0

6 Sabotage

No-Go 6.1 (Surrogate models; CH06). A piecewise-linear analog forecaster built on 20000 consecutive pairs has one-step error below 10^{-12} . A protocol certifying it at all 200 horizons on the strength of that fit holds only 17 steps on held-out trajectories at $\sigma_0 = 10^{-6}$, which is exactly the frontier. That leaves 183 false LOCKs. RG certifies only up to the measured frontier.

No-Go 6.2 (Static gates are chaos-blind; CH07). An equal-time inversion gate (noisy linear sensors to state) passes on a chaotic map (NRMSE 0.002) and on a periodic map (0.011) alike. Its frontiers are 17 and ∞ . Only a gate whose decision depends on the dynamics can separate them. This is the defect of the chaos/coil attack.

Remark 6.3 (Canaries). Two canaries are run:

- *Task substitution*: silently re-observing the state each step makes the chaotic frontier vanish. CH01 rejects this as hidden chaos.
- *Absorption*: declaring the protocol slopes different also fails its check.

Both are caught.

7 What this means for RG

RG's statuses are not a filter laid over the dynamics; they are a measuring instrument:

- the boundary between LOCKED and ESCALATE measures λ ;
- its dependence on the coverage level measures the tail of the finite-time exponents;
- the growth of the quotient measures h ;
- the Fisher information about the past measures 2λ .

Each sits inside an object the ladder already defines: certificates (L38, L68), the kernel of the history protocol (L66), and the Fisher module (L22). A declaration could hide chaos only by changing these slopes. By the frontier law, no protocol can do that without changing the task, and a change of task is a declaration RG records. In this sense chaos is carried by construction: it is conserved between the declared precision and the certified horizon, and it is never silently discharged.

8 Limits

- The slopes are asymptotic. At finite precision they are window certificates (L38). Near $\lambda = 0$, as at the Feigenbaum point, the frontier slope diverges and certificates abstain.
- Only the top exponent shows in a state forecast. The full Lyapunov spectrum needs volume forecasts, that is, forecasts of k -dimensional patches.
- The Sabra coil system was not rerun here because of its cost. The frontier law applies to it, and a dynamic-gate version of that attack is the natural next battery.
- Generating partitions are assumed for the entropy statement. Coarser partitions can under-report h , and that is a resolution statement (window), not a loss of chaos.

9 Place in the ladder

- The frontier law and the duality refine L65–L69 (dynamics) and connect them to L38 (window certificates) and L22 (Fisher).
- The conservation corollary is an instance of maximal closure (L87–L90): chaos appears as ESCALATE or STRUCTURAL WALL, never as a false LOCKED.

A Reproducibility

`rg_chaos_frontier_attack.py` passes 7/7 on seeds 20261007, 1, 42, 777 and 2026; `--figs` writes the figures, and `--canary` runs the two canaries. Corrections before registration:

1. CH01 first required the 95% slope to equal λ for every system and failed for $r = 3.9$ and Hénon. The tail of the finite-time exponents explains this, and the check was re-registered on that mechanism.
2. The periodic control first included $\sigma_0 = 10^{-2}$, where a finite transient exceeds ε .
3. The first task-substitution canary crashed instead of failing; a vanished frontier is now a registered rejection.

References

- [1] V. I. Oseledets, A multiplicative ergodic theorem, *Trans. Moscow Math. Soc.* 19 (1968), 197–231.
- [2] A. Katok, B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*, Cambridge, 1995.
- [3] M. Misiurewicz, W. Szlenk, Entropy of piecewise monotone mappings, *Studia Math.* 67 (1980), 45–63.
- [4] P. Walters, *An Introduction to Ergodic Theory*, Springer, 1982.
- [5] M. Rada, *Resolution Geometry: The Theorem Ladder*, 2026.
- [6] M. Rada, *Resolution Geometry on Chaotic and Dirty Experiments*, 2026.